Dhruv Athaide

Application Security Engineer · Mobile / Product Security · VAPT & Red Team · Published Researcher

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SUMMARY

Application Security Engineer with 2+ years securing production mobile software – mobile Vulnerability Assessment & Penetration Testing (VAPT), threat modeling, applied cryptography, and secure SDLC ownership across **5 apps shipped to the Google Play Store**. Hands-on with OWASP MASVS/MASTG, static & dynamic analysis (SAST/DAST), reverse engineering, and AOSP hardening (NDK/JNI, SELinux). Co-author of a peer-reviewed IGI Global book chapter (2026) on Android vulnerability assessment.

TECHNICAL SKILLS

Security & AppSec: VAPT, OWASP MASVS/MASTG & Mobile Top 10, Threat Modeling (STRIDE), Secure SDLC (SSDLC), Secure Code Review, Static & Dynamic Analysis (SAST/DAST), CWE/CVSS, Software Composition Analysis (SCA), AOSP/SELinux hardening, JNI/NDK anti-tamper

Cryptography: AES-256-GCM, ChaCha20-Poly1305, X3DH & Double Ratchet (Signal), end-to-end encryption (E2EE), ECDSA/Ed25519, SQLCipher, hardware-backed keystore

Tooling: Burp Suite, Frida, Ghidra, Nmap, Wireshark, Git, CI/CD, Docker, Azure

Languages & Platforms: Kotlin, Java, C++ (NDK/JNI), Python, TypeScript, SQL; Android SDK, Jetpack Compose, Flutter, Node.js, Firebase, MongoDB

EXPERIENCE

Cyber Analyst – Application & Mobile Security — Mumbai, India

DeepCytes Cyber Labs UK — Aug 2025 – Present

- Own mobile application security end-to-end – threat modeling, secure code review, pre-release security gating, and native anti-tamper hardening – across a product line of 5 apps shipped to Google Play.
- Conduct mobile VAPT and secure-design reviews against OWASP MASVS/MASTG, identifying access-control, insecure-storage, and crypto-misuse class findings (CWE-mapped) and driving remediation to closure on retest.
- Build offensive Android tooling for red team engagements mapped to the MITRE ATT&CK Mobile framework (covert telemetry, permission analysis, threat intelligence), and harden an AOSP distribution with custom kernel modifications for high-assurance (HNI / government) builds.
- Define secure-SDLC practice (code review, dependency/SCA checks, release gating) and mentor junior engineers across the mobile line.

Red Team & Mobile Dev Team Lead (Internship) — Mumbai, India

DeepCytes Cyber Labs UK — Jul 2024 – Jun 2025

- Led a cross-functional team across penetration testing, threat intelligence, and vulnerability research; delivered four enterprise platforms (incl. an LMS and a SaaS product) and drove the initial architecture of the Torch remote-wipe platform.

KEY PROJECTS

Guardient – Mobile Privacy & Threat Intelligence (Play Store)

Kotlin, Android SDK, C++/NDK, Jetpack Compose, MVVM, Room

- Architected a no-root per-app firewall using Android's `VpnService` TUN interface, black-holing unauthorized app traffic at the kernel routing-table level with no throughput penalty.
- Built a deterministic permission-risk engine for toxic combinations (banking-trojan overlay, 2FA scrapers), an on-device phishing/homoglyph URL analyzer, and a quantized on-device LLM assistant (MediaPipe) – fully offline.

NoSurveil – BLE Anti-Stalking & Tracker Detection (Play Store)

Kotlin, C++/NDK, Room + SQLCipher, Hilt, BLE/GATT

- Engineered a counter-surveillance pipeline classifying 23 device types across BLE, Wi-Fi, and ASTM F3411 drone Remote-ID, with custom Apple Continuity and Google Fast Pair parsers.
- Designed a following-detection algorithm (trimmed-mean bucketing, Haversine vectors, RSSI fade thresholds) separating shadowing from stationary beacons, with RPA re-linking across rotating MACs.

LockGuard+ – Intruder Detection & Forensic Evidence (Play Store)

Kotlin, CameraX, C++/NDK, AES-256, WorkManager, Play Billing

- Built a headless CameraX capture pipeline via a synthetic `LifecycleOwner`, recording intruder photo/video on failed unlocks while the device is locked, Android 7.1–16.
- Hardened against Frida/Xposed hooks and APK repackaging via a `BiometricPrompt.CryptoObject` probe and a native C++ layer with XOR-fragmented secrets and runtime package attestation; shipped with Play Billing.

SecureLine – Zero-Trust E2EE Messaging & Calling (Play Store)

Kotlin, Compose, Node.js/Fastify, WebRTC, Tor, SQLCipher, MongoDB

- Implemented the Signal protocol (X3DH + Double Ratchet over Curve25519) with hardware-backed Ed25519 keys and a CONIKS-style key-transparency verifier over append-only SHA-256 chains.
- Built WebRTC mesh voice/video with sealed signaling and SAS MITM detection, and embedded an on-device C-Tor daemon hosting ephemeral v3 onion chat rooms.

EDUCATION

Indian Institute of Technology Patna — Aug 2026 – Jul 2028

M.S. in Computer Science & Data Analytics

Amity University Mumbai — Jul 2022 – Jun 2025

Bachelor of Computer Applications (BCA) — **CGPA: 9.69 / 10.0**

RESEARCH & CERTIFICATIONS

Publication: *Design of Comprehensive Mobile Device Vulnerability Checking Application* – Book Ch. 6, IGI Global, 2026. DOI · ORCID

Certifications: Google Cloud Computing Foundations (Credly) · Google Developer Student Club Web Developer